

ALAN GEBHARDTSBAUER

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Site Reliability Engineer — CI/CD, Observability & Distributed Systems

Engineer with a decade of experience hardening CI/CD pipelines, resolving concurrency and reliability issues in production data systems, and building observability into multi-service async pipelines on AWS.

Technical Stack & Proficiencies

- **AWS:** EventBridge, CloudWatch, Lambda, S3, Kinesis Firehose, AWS-SDK
- **CI/CD:** GitLab CI, Docker, sorry-cypress, build-test-deploy pipelines
- **Data:** SQL, Snowflake, DynamoDB, Mongo
- **Systems:** Bash, Linux (Arch, Nix, Alpine, Debian), concurrency/locking
- **Observability:** CloudWatch, custom reporting/alerting pipelines
- **Languages:** TypeScript/Node, SQL, Bash

Career Experience

Centene

Arlington, Virginia -> Remote

Lead Applications Software Engineer in Test

February 2020 – Present

- Led test automation efforts across Centene's web platform, directly guiding 4-5 engineers at a time (dozens over 6 years) and running recurring technical brown-bag sessions on Git workflows and test tooling.
- Built a self-healing agentic test-maintenance pipeline (Playwright MCP) that reduces manual upkeep as UI changes, automatically routing unresolvable failures to human review instead of blocking silently.
- Built XP_CYTEST's CI layer: templated, configurable GitLab jobs (run mode, browsers, parallelization, accessibility scanning) plus automated Microsoft Teams reporting, self-service across 15+ teams (and growing) on the platform scaffold.
- Architected a Playwright test-scaffolding CLI with per-domain environment generation and cached authenticated sessions, scaling parallel FE test execution across 28+ brand domains.
- Overhauled org-wide Cypress CI/CD pipelines: introduced configurable Cypress versioning, pre-built Docker images (cutting ~5 min/job), and sorry-cypress parallelization to eliminate recurring 1-hour runner timeouts; led migration of tests from a standalone repo into application repos, eliminating version skew.
- Automated triggering and polling of an asynchronous, multi-system batch pipeline (web portal -> MongoDB -> Linux batch job -> PDF generation), with state verification at every stage, replacing a fully manual, serialized process.
- Resolved a concurrency bottleneck in a legacy, single-threaded SQL stored-procedure engine by scoping batch processing to specific records, enabling parallel testing in place of exclusive locks; built a reversible test-scenario system constraining data mutation to sanctioned, undoable operations.
- Built a CLI orchestrator automating end-to-end verification of an asynchronous, multi-service data pipeline (S3 -> Genesys/DynamoDB -> CloudWatch -> Kinesis Firehose -> Snowflake), confirming delivery integrity and observability across every hop.
- Automated Jira ticket-status transitions from GitLab CI events (branch created, MR opened, deploy stages), keeping ticket state accurate without manual updates.

FINRA

Rockville, Maryland

Software Development Engineer in Test

April 2018 – January 2020

- Automated end-to-end testing using Java, Selenium, JDBC, and JUnit.
- Automated SQL regression testing validating data integrity through CRUD operations.
- Wrote and executed end-to-end test cases and documented application issues in Jira.

Harmonia Holdings LLC.

Blacksburg, Virginia

Quality Assurance Engineer

March 2016 – March 2018

- Acquired Public Trust Clearance with the Department of Agriculture.
- Automated testing using Java, Selenium, SQL, JMeter, and QTest.

Mountain State Information Systems

Radford, Virginia

Software Engineering Intern

June 2015 – March 2016

- Web app development and maintenance: HTML, CSS, JavaScript, PHP, FTP.
- Automated cost comparisons and visualized data with Visual Basic macros.

Projects

- **testwright** — Generalized an internal multi-domain test-infrastructure platform (originally scaled to 28+ brand domains) into a published npm package — CLI-scaffolded environments, cached session management, and CI wiring as reusable platform tooling rather than one-off scripts.
- **agentic-testing** — Personal rebuild of a self-healing pipeline (requirements -> test generation -> execution -> self-healing -> reporting) — pipeline-as-platform design with automated failure recovery and human escalation on unresolved cases.
- **mousehole** — Native Zig + GTK4 Linux tool combining Wayland screenshot capture, text parsing, and window-manager (hyprctl) integration to drive precise cursor control.

Education

B.S., Computer Science – Radford University – GPA: 3.73

Graduated December 2015